

SmartBranch 360 Network Design

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Problem Statement:

SmartBranch 360 Network Design Contribution: As a solo participant, I independently designed and implemented the entire network architecture in Cisco Packet Tracer. My contributions include wiring the physical topology, configuring Inter-VLAN routing (Router-on-a-Stick) for Employee (VLAN 10), Guest (VLAN 20), Server (VLAN 30), and Management (VLAN 99) networks, establishing DHCP pools, and implementing strict security ACLs to isolate Guest traffic and secure Admin SSH access. I also developed a Python script to automate the validation of the router's interface configurations.

Fault Scenario 1: Missing Trunk VLAN

- **Symptom:** Guest laptops have no internet or network connectivity.
- **Root Cause:** VLAN 20 was missing from the allowed list on the trunk link between SW1 and SW2.
- **Fix:** Accessed SW1 CLI, entered "interface gigabitEthernet 0/2", and applied the command "switchport trunk allowed vlan add 20"

Fault Scenario 2: Incorrect Default Gateway

- **Symptom:** Employee PCs cannot reach the Internal Server.
- **Root Cause:** The Internal Server's static IP configuration had a typo in the Default Gateway (192.168.30.100 instead of 192.168.30.1).
- **Fix:** Opened Server IP Configuration tab and manually corrected the Default Gateway to "192.168.30.1".

Fault Scenario 3: Broken DHCP Pool

- **Symptom:** Employee PCs auto-assign 169.254.x.x (APIPA) addresses instead of 192.168.10.x.
- **Root Cause:** The router's DHCP pool for VLAN 10 was configured with the wrong network subnet mask.
- **Fix:** Accessed R1, removed the bad pool, and recreated it with network "192.168.10.0 255.255.255.0" and "default-router 192.168.10.1".

Fault Scenario 4: Overly Restrictive Security ACL

- **Symptom**: The Management PC (PC4) gets a "Connection Refused" error when attempting to SSH into the router.
- **Root Cause**: The "ADMIN_ONLY" standard ACL was missing the permit statement for the 192.168.99.0 network.
- **Fix**: Accessed R1 CLI, entered the ACL configuration, and added "permit 192.168.99.0 0.0.0.255".

Fault Scenario 5: Administratively Down Sub-Interface

- **Symptom**: The entire Server VLAN (VLAN 30) is unreachable from any other network segment.
- **Root Cause**: The router sub-interface "GigabitEthernet0/0/0.30" was in an "administratively down" state.
- **Fix**: Accessed R1 CLI, entered interface GigabitEthernet0/0/0.30, and executed the "no shutdown" command.

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validate.py ×

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```
1 # SmartBranch 360 - Network Configuration Validation Tool
2 # Author: Srikanth Paleti
3
4 print("--- Starting SmartBranch 360 Configuration Check ---\n")
5
6 # 1. Define the Expected Network Plan
7 expected_vlans: list[str] = ["10", "20", "30", "99"]
8 expected_gateways: dict[str, str] = {
9     "10": "192.168.10.1",
10    "20": "192.168.20.1",
11    "30": "192.168.30.1",
12    "99": "192.168.99.1"
13 }
14
15 # 2. Simulated 'show ip interface brief' output from R1
16 router_show_output = """
17 Interface                IP-Address    OK? Method Status          Protocol
18 GigabitEthernet0/0/0     unassigned    YES unset  up              up
19 GigabitEthernet0/0/0.10  192.168.10.1  YES manual up              up
20 GigabitEthernet0/0/0.20  192.168.20.1  YES manual up              up
21 GigabitEthernet0/0/0.30  192.168.30.1  YES manual up              up
22 GigabitEthernet0/0/0.99  192.168.99.1  YES manual up              up
23 """
24
25 # 3. Validation Logic
26 error_count = 0
27
28 for vlan in expected_vlans:
29     target_interface: str = f"GigabitEthernet0/0/0.{vlan}"
30     target_ip: str = expected_gateways[vlan]
31
32     # Check if the interface and correct IP exist in the router output
33     if target_interface in router_show_output and target_ip in router_show_output:
34         print(f"[PASS] VLAN {vlan} Gateway ({target_ip}) is active and configured correctly.")
35     else:
36         print(f"[FAIL] Configuration error detected on VLAN {vlan}!")
37         print(f"    Suggested Fix: Check sub-interface {target_interface} or IP assignment.")
38         error_count += 1
39
40 print(f"\n--- Validation Complete. Errors Found: {error_count} ---")
```

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- flinty@M2-Air-15 ~ % cd ~/Downloads
- flinty@M2-Air-15 Downloads % python3 validate.py

```
--- Starting SmartBranch 360 Configuration Check ---

[PASS] VLAN 10 Gateway (192.168.10.1) is active and configured correctly.
[PASS] VLAN 20 Gateway (192.168.20.1) is active and configured correctly.
[PASS] VLAN 30 Gateway (192.168.30.1) is active and configured correctly.
[PASS] VLAN 99 Gateway (192.168.99.1) is active and configured correctly.

--- Validation Complete. Errors Found: 0 ---
flinty@M2-Air-15 Downloads %
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